

Paper Title

1st Given Name Surname
*Department of Electrical and
Information Engineering
Universitas Gadjah Mada
Yogyakarta, Indonesia
email address or ORCID*

2nd Given Name Surname
*Department of Electrical and
Information Engineering
Universitas Gadjah Mada
Yogyakarta, Indonesia
email address or ORCID*

3rd Given Name Surname
*Department of Electrical and
Information Engineering
Universitas Gadjah Mada
Yogyakarta, Indonesia
email address or ORCID*

Abstract—This document is a model and instructions for L^AT_EX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

The continuous growth of Indonesia’s population has driven a steady increase in household electricity demand. According to Statistics Indonesia (BPS), the national population growth rate has consistently ranged between 1% and 1.5% per year [1], while data from the National Socioeconomic Survey (SUSENAS) confirm a corresponding rise in household electronic appliance ownership [2]. As households increasingly rely on devices such as air conditioners, televisions, and refrigerators, total residential electricity consumption continues to grow. Nevertheless, public understanding of efficient energy consumption patterns remains limited [3].

A key barrier to energy-efficient behavior is the lack of accessible, actionable feedback on household energy use. Most users only receive aggregated information through monthly electricity bills, without understanding which appliances contribute most to their total consumption or how house design affects energy usage. Studies show that static, non-personalized energy feedback is insufficient to stimulate meaningful behavioral change [4]. Interactive and visual approaches have been shown to be more effective in raising energy literacy among building occupants [5], and gamification or simulation-based methods have demonstrated potential in motivating energy behavior change [6].

Virtual Reality (VR) has emerged as a powerful medium for interactive visualization, particularly in the architecture, engineering, and construction sectors [7]. Prior work has demonstrated that VR enables users to explore and customize house designs in an immersive environment [8]. However, existing VR applications for residential design have focused primarily on spatial visualization and aesthetic customization, without addressing electricity consumption simulation or energy estimation features. The integration of energy awareness into VR-based house design tools remains largely unexplored.

This paper presents *ModXplore*, an interactive VR-based simulation system developed for Meta Quest 3 using the Unity Engine. ModXplore allows users to explore and customize a

virtual house while simultaneously simulating and estimating household electricity consumption using the Bottom-Up Appliance Modeling approach. The system further integrates a Large Language Model (LLM)—specifically LLaMA 3.1 via the Groq API—to automatically generate personalized appliance usage schedules. The system was evaluated through black-box functional testing and a user study with 24 respondents using the System Usability Scale (SUS), User Experience Questionnaire (UEQ), and a Likert-scale questionnaire on energy understanding and decision-making support.

The main contributions of this work are: (1) an interactive VR simulation system integrating house design exploration with real-time electricity consumption estimation; (2) an energy estimation module based on Bottom-Up Appliance Modeling that accurately distinguishes usage scenarios; and (3) an AI-powered schedule personalization feature using an LLM to assist users in making energy-efficient decisions.

II. RELATED WORK

A. VR for Building Visualization and Design

VR has been widely adopted in architecture and residential design to overcome the limitations of conventional 2D visualization. Lang and Zhuang demonstrated that combining VR with clustering algorithms for interior design improved modeling accuracy by over 20% compared to traditional SVM-based methods and enabled real-time evaluation of design changes [9]. Safikhani et al. confirmed that immersive VR environments significantly enhance users’ spatial understanding by allowing first-person exploration of buildings, which is difficult to achieve with static renders [7]. Wang et al. further showed that scene-context-aware object interaction in VR reduces selection time and raises user satisfaction during interior configuration tasks [10]. The integration of VR with Building Information Modeling (BIM) has also been shown to improve simulation realism and design communication between users and architects [11], [12]. In the Indonesian context, Suryanto developed a VR application for house design exploration and material customization on Meta Quest 3 using Unity, achieving a SUS score of 70.63; however, the system did not include any electricity consumption simulation or energy estimation feature [8].

B. VR and AR for Energy Simulation and Awareness

Beyond architectural visualization, VR and AR are increasingly applied to improve energy literacy and support energy-efficient behavior. Latini et al. developed an immersive VR training environment that integrates a residential Building Energy Model with real-time eco-feedback indicators; over 90% of participants showed improved knowledge of energy-related behavior after the session [13]. Maqbool et al. evaluated audio, visual, and social cues in VR for promoting energy conservation, finding that visual interventions produced statistically significant improvements in pro-environmental behavior scores [14]. Anifowose et al. introduced ENERGYSIM, a multi-user immersive VR prototype for building energy education that allows users to explore the impact of building materials on energy consumption through interactive menu-driven walkthroughs [15]. In the AR domain, An et al. showed that overlaying real-time energy consumption data onto physical appliances via AR reduced energy use without degrading occupant comfort [16]. Frigura-Iliasa et al. proposed an interactive 3D simulator (ElectroHouse) connecting consumption graphs with appliance-level controls to improve household energy literacy [17].

Despite this progress, existing studies either focus on building-level energy modeling without appliance-level interactivity, or on energy awareness without integrating house design exploration. None of the reviewed systems combine immersive VR house customization, bottom-up appliance-level energy estimation, and AI-driven schedule personalization within a single platform, which is the gap this work addresses.

III. PREPARE YOUR PAPER BEFORE STYLING

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections III-A–III-E below for more information on proofreading, spelling and grammar.

Keep your text and graphic files separate until after the text has been formatted and styled. Do not number text heads— $\LaTeX$ will do that for you.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance

dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.

- Do not mix complete spellings and abbreviations of units: “Wb/m²” or “webers per square meter”, not “webers/m²”. Spell out units when they appear in text: “. . . a few henries”, not “. . . a few H”.
- Use a zero before decimal points: “0.25”, not “.25”. Use “cm³”, not “cc”.)

C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \quad (1)$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not “Eq. (1)” or “equation (1)”, except at the beginning of a sentence: “Equation (1) is . . .”

D. $\LaTeX$ -Specific Advice

Please use “soft” (e.g., `\eqref{Eq}`) cross references instead of “hard” references (e.g., (1)). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don’t use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

Please note that the `{subequations}` environment in $\LaTeX$ will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you’ve discovered a new method of counting.

$\BibTeX$ does not work by magic. It doesn’t get the bibliographic data from thin air but from .bib files. If you use $\BibTeX$ to produce a bibliography you must send the .bib files.

$\LaTeX$ can’t read your mind. If you assign the same label to a subsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

$\LaTeX$ does not have precognitive abilities. If you put a `\label` command before the command that updates the counter it’s supposed to be using, the label will pick up the last counter to be cross referenced instead. In particular, a `\label` command should not go before the caption of a figure or a table.

Do not use `\nonumber` inside the `{array}` environment. It will not stop equation numbers inside `{array}` (there won’t be any anyway) and it might stop a wanted equation number in the surrounding equation.

E. Some Common Mistakes

- The word “data” is plural, not singular.
- The subscript for the permeability of vacuum μ_0 , and other common scientific constants, is zero with subscript formatting, not a lowercase letter “o”.
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an “inset”, not an “insert”. The word alternatively is preferred to the word “alternately” (unless you really mean something that alternates).
- Do not use the word “essentially” to mean “approximately” or “effectively”.
- In your paper title, if the words “that uses” can accurately replace the word “using”, capitalize the “u”; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones “affect” and “effect”, “complement” and “compliment”, “discreet” and “discrete”, “principal” and “principle”.
- Do not confuse “imply” and “infer”.
- The prefix “non” is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the “et” in the Latin abbreviation “et al.”.
- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

An excellent style manual for science writers is [?].

F. Authors and Affiliations

The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

G. Identify the Headings

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is “Heading 5”. Use “figure caption” for your Figure captions, and “table head” for your table title. Run-in heads, such as “Abstract”, will require you

to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

H. Figures and Tables

a) *Positioning Figures and Tables:* Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 1”, even at the beginning of a sentence.

TABLE I
TABLE TYPE STYLES

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
copy	More table copy ^a		

^aSample of a Table footnote.



Fig. 1. Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

REFERENCES

Please cite all your references [18], [19]. References are stored in a bibtex file "references.bib". You can use Mendeley or Jabref for your reference manager.

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